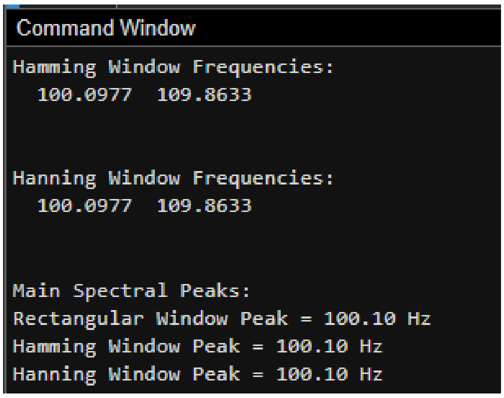
7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

CODE:

```
clc; clear; close all;
% Signal
fs = 1000; N = 256; t = (0:N-1)/fs;
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);
% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hanning'};
% Processing
for i = 1:3
   xw = x(:).*w{i};
   X = fft(xw,NFFT);
   P{i} = abs(X(1:NFFT/2));
   P{i} = P{i}/max(P{i});
end
%% Individual Spectrums
figure;
for i = 1:3
   subplot(3,1,i);
   plot(f,20*log10(P{i}),'LineWidth',1.5);
   title(['Spectrum using ' names{i} ' Window']);
   xlim([0 250]); grid on;
end
%% Comparison
figure;
plot(f,20*log10(P{1}),'k',f,20*log10(P{2}),'r',f,20*log10(P{3}),'b','LineWidth',1.5);
legend(names); title('Window Comparison'); xlim([50 170]); grid on;
%% Window Shapes
```

```
figure;
for i = 1:3
    subplot(3,1,i); plot(w{i},'LineWidth',1.5);
    title([names{i} ' Window']); grid on;
end
%% Peak Detection
for i = 1:3
    [pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);
    fprintf('\n%s Window Frequencies:\n',names{i});
    disp(loc{i});
end
%% Peak Marking (Hamming)
figure;
plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;
plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');
title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;
%% Main Peaks
fprintf('\nMain Spectral Peaks:\n');
for i = 1:3
    [~,idx] = max(P{i});
    fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));
end
```

OUTPUT:



```
Command Window
Hamming Window Frequencies:
  100.0977  109.8633


Hanning Window Frequencies:
  100.0977  109.8633


Main Spectral Peaks:
Rectangular Window Peak = 100.10 Hz
Hamming Window Peak = 100.10 Hz
Hanning Window Peak = 100.10 Hz
```

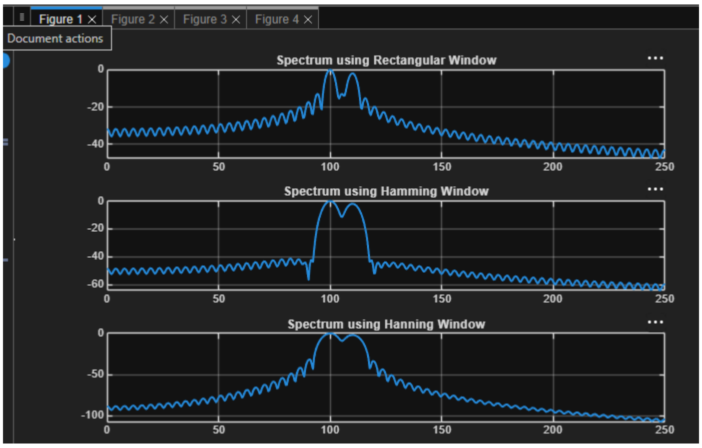


Figure 1
Figure 2
Figure 3
Figure 4
Document actions
Spectrum using Rectangular Window
0
-20
-40
0
50
100
150
200
250
Spectrum using Hamming Window
0
-20
-40
-60
0
50
100
150
200
250
Spectrum using Hanning Window
0
-50
-100
0
50
100
150
200
250

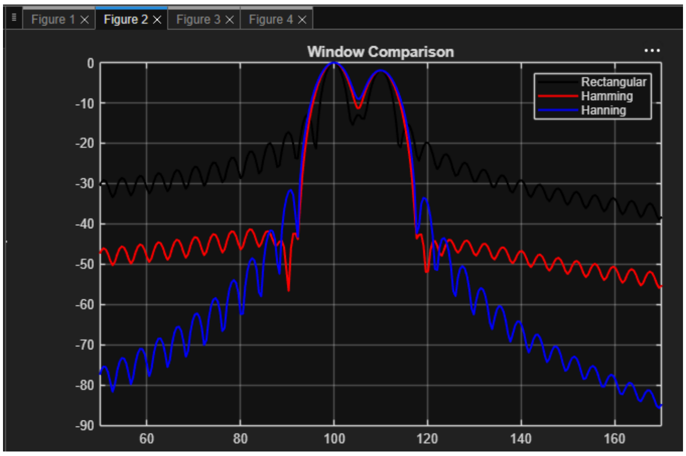


Figure 1
Figure 2
Figure 3
Figure 4
Window Comparison
Rectangular
Hamming
Hanning
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
60
80
100
120
140
160

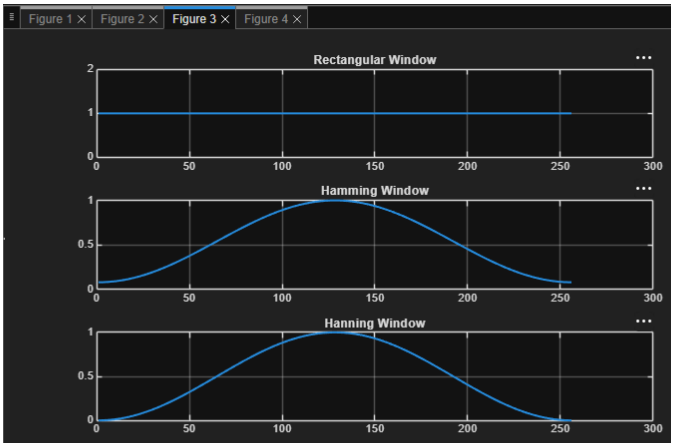


Figure 1
Figure 2
Figure 3
Figure 4
Rectangular Window
2
1
0
0
50
100
150
200
250
300
Hamming Window
1
0.5
0
0
50
100
150
200
250
300
Hanning Window
1
0.5
0
0
50
100
150
200
250
300

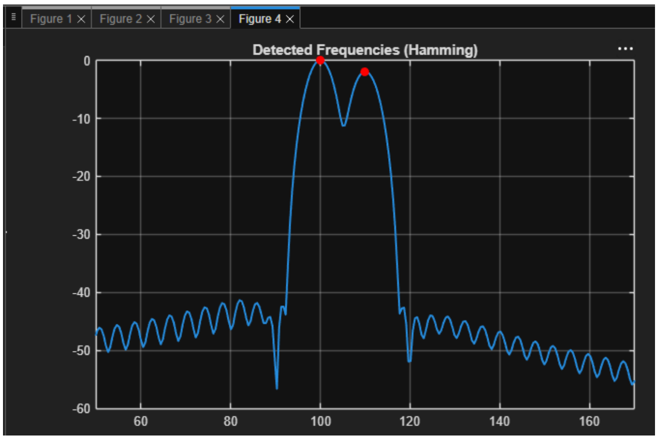


Figure 1
Figure 2
Figure 3
Figure 4
Detected Frequencies (Hamming)
0
-10
-20
-30
-40
-50
-60
60
80
100
120
140
160